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Comprehensive Genomic Profiling Landscape and the Clinical Impact of BRCA 1/2 Pathogenic Variants in the Metastatic Castration-Resistant Prostate Cancer

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Background

- ✓Metastatic castration-resistant prostate cancer (mCRPC) shows marked clinical and genomic heterogeneity.
- ✓Alterations in homologous recombination repair (HRR) genes, particularly *BRCA1/2*, are clinically actionable.
- ✓PARP inhibitor olaparib has demonstrated survival benefit in *BRCA*-altered mCRPC (PROfound trial).
- ✓Real-world data evaluating genomic heterogeneity and variant-specific outcomes remain limited, especially in Asian populations.
- ✓This study aimed to characterize the genomic landscape of HRR alterations and evaluate the prognostic and therapeutic impact of *BRCA1/2* pathogenic variants in a nationwide Japanese cohort.

Methods

- ✓Retrospective cohort study using the **C-CAT national precision oncology database** in Japan.
- ✓Included **5,893 patients with mCRPC** who underwent comprehensive genomic profiling (CGP).
- ✓Five reimbursed CGP platforms were analyzed (tumor tissue and liquid biopsy).
- ✓HRR alterations defined based on pathogenic or likely pathogenic annotations (**12 HRR genes including *BRCA1, BRCA2, ATM, CDK12, CHEK2, PALB2, RAD51C, RAD51D, BRIP1, BARD1, FANCA*, and *FANCL***).
- ✓**Primary endpoints:** Overall survival after olaparib initiation (OS_olaparib)
- ✓Survival analyses performed using Kaplan–Meier and multivariable Cox regression models.

Results/Graphs/Data

Figure 1. Patient characteristics in total cohort

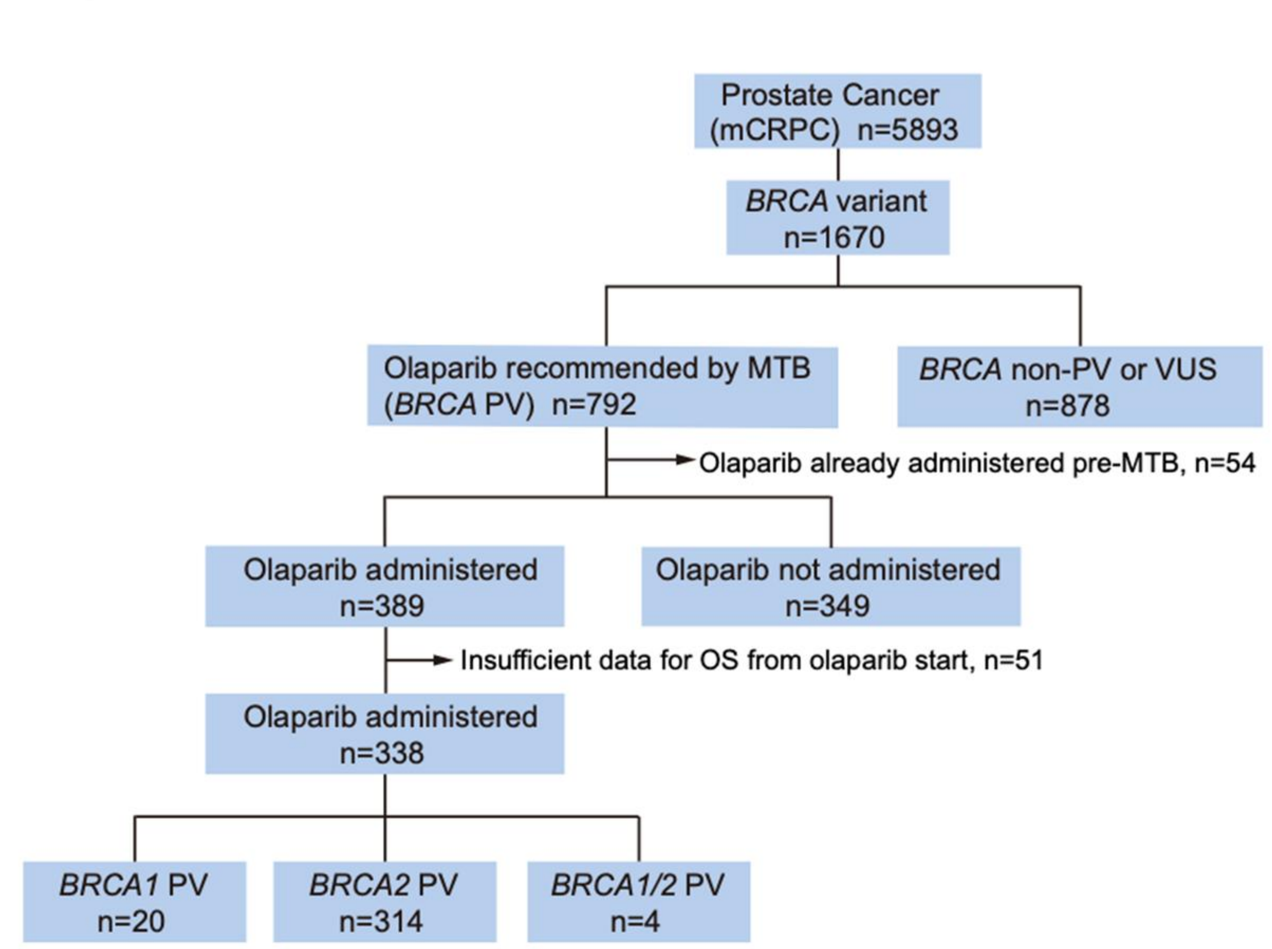
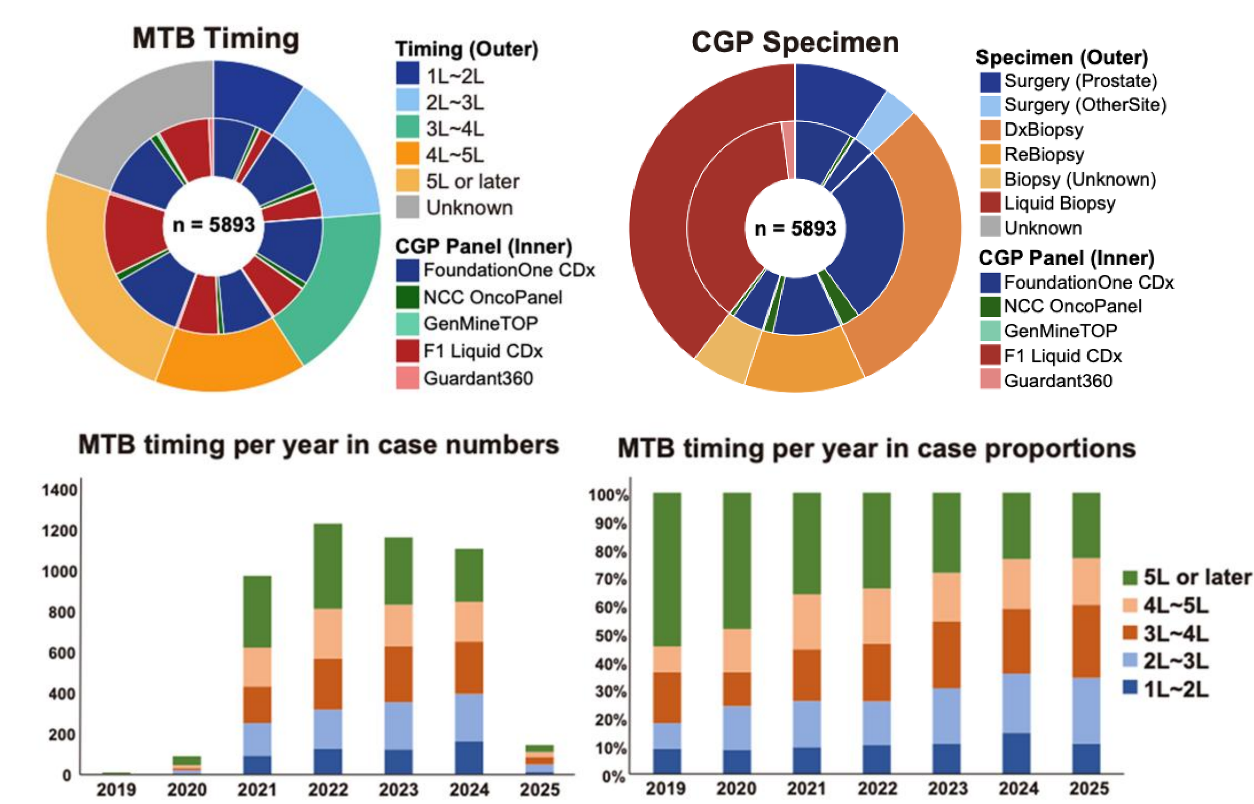
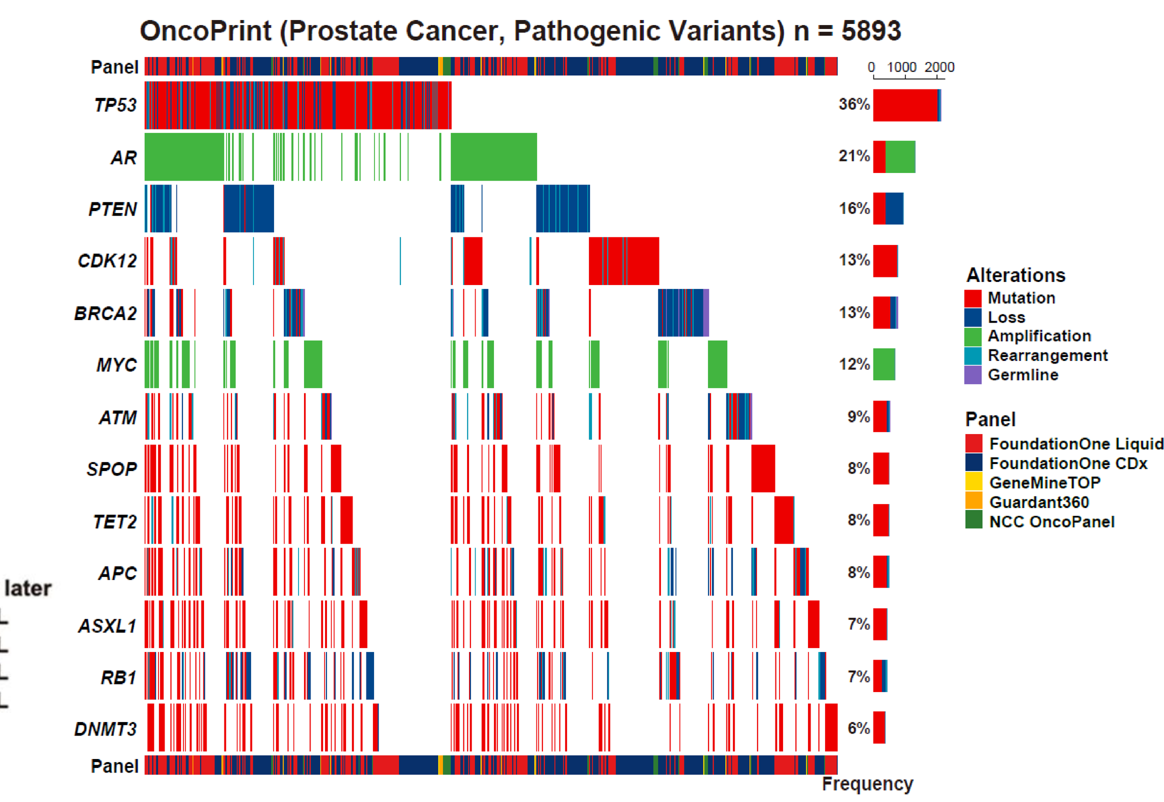


Figure 2. MTB timing and CGP specimen



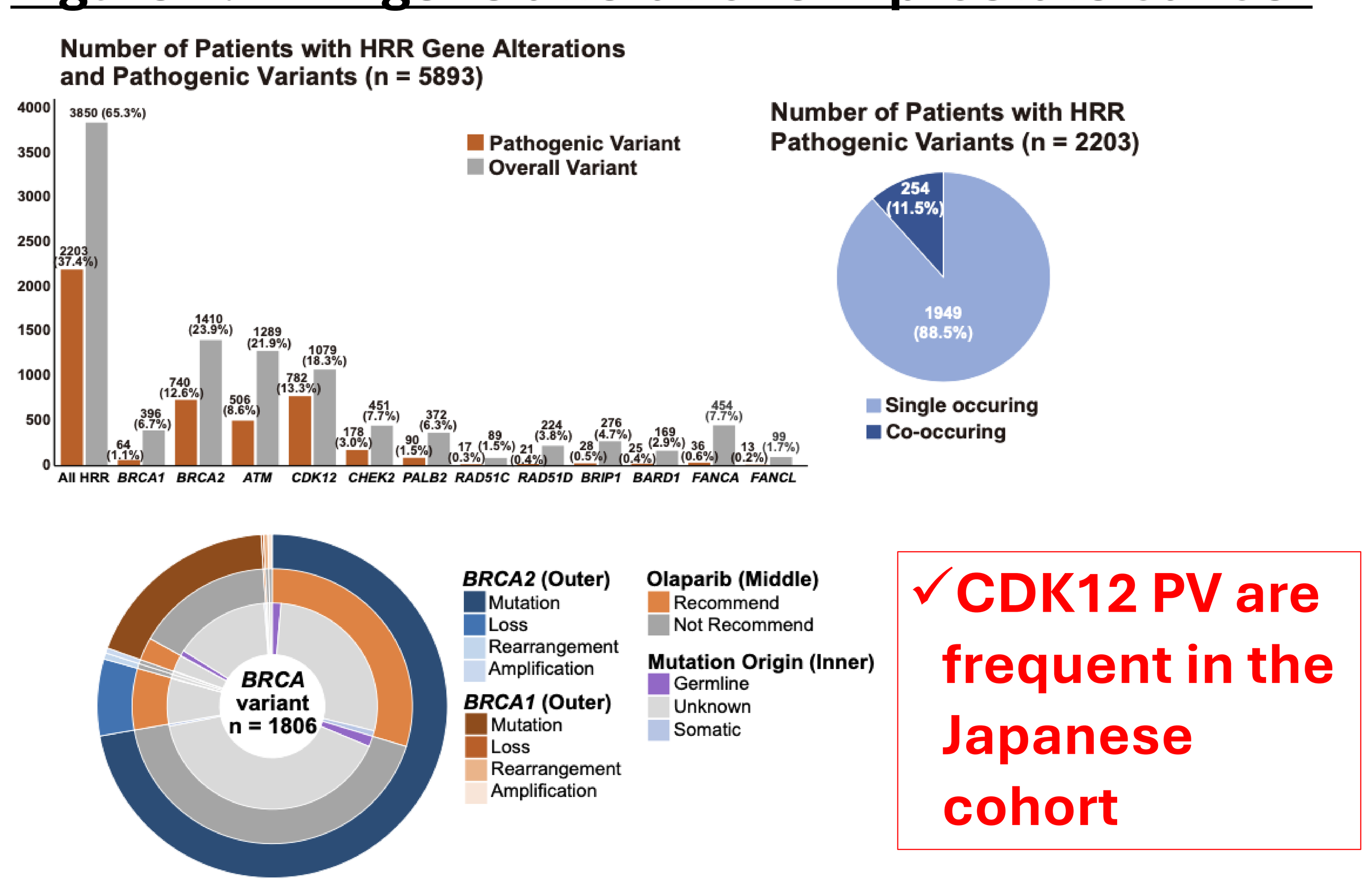
✓ A gradual trend toward earlier MTB review was seen in recent cases

Figure 3. Genomic landscape in mCRPC



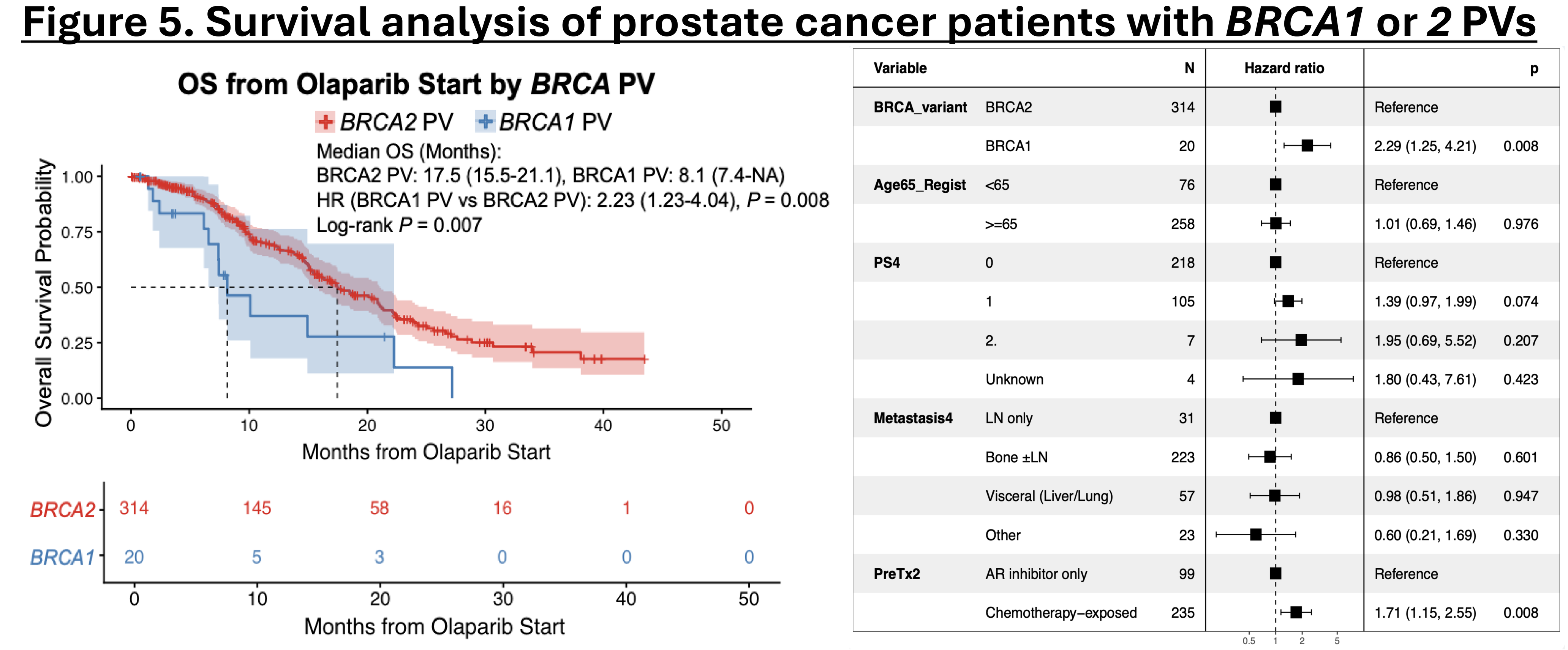
✓PVs were most frequent in TP53, followed by BRCA2, CDK12, and ATM .

Figure 4. HRR gene alterations in prostate cancer



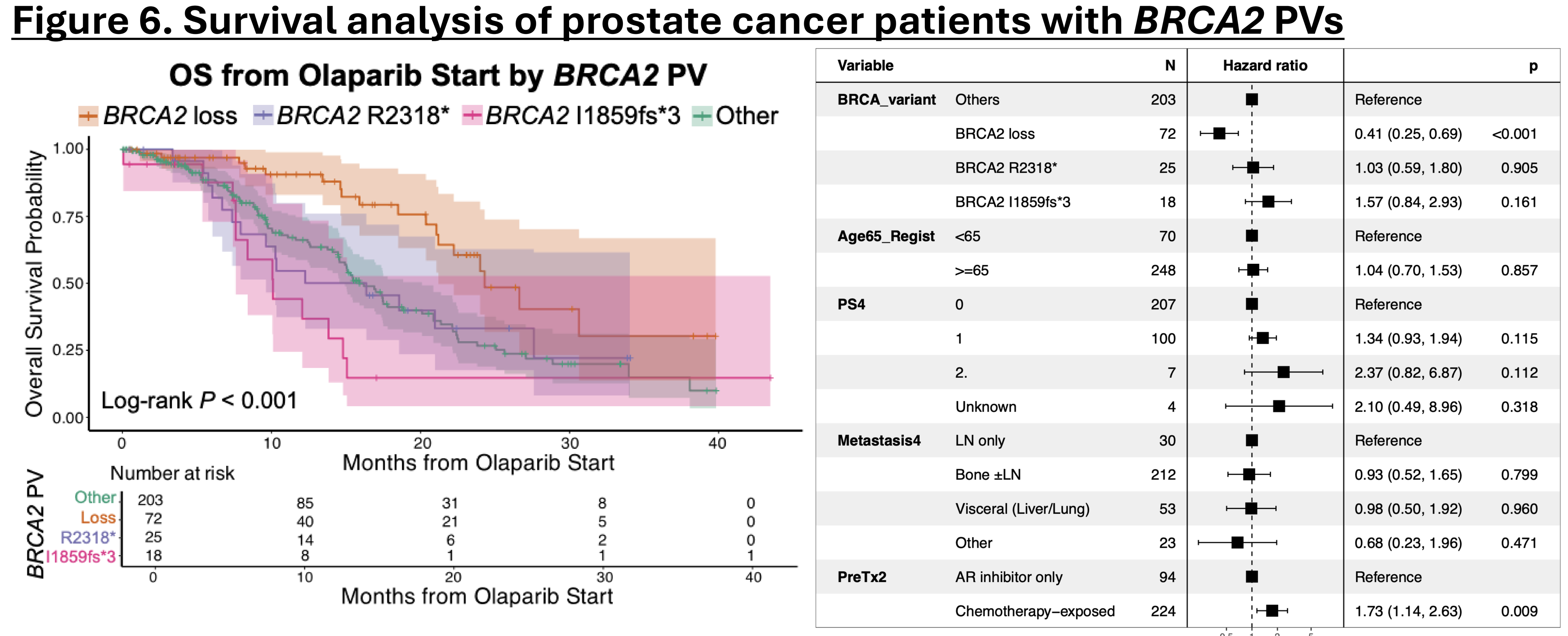
✓CDK12 PV are frequent in the Japanese cohort

Figure 5. Survival analysis of prostate cancer patients with *BRCA1* or *2* PVs



✓*BRCA1* PVs had inferior outcomes than those with *BRCA2* PVs

Figure 6. Survival analysis of prostate cancer patients with *BRCA2* PVs



✓*BRCA2* loss had the most favorable response to olaparib

Conclusion

Variant-level annotation of *BRCA* alterations offers clinically meaningful prognostic and predictive insights beyond *BRCA* mutation status alone in patients with mCRPC receiving olaparib.